

Migrations

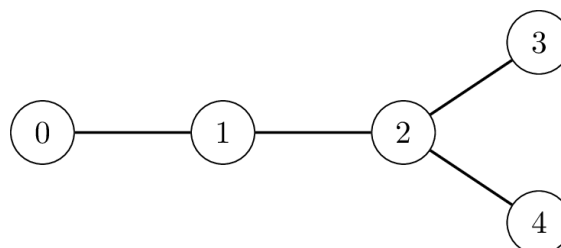
Tekst zadatka je doslovno preveden iz izvorne verzije. Čitanje nije preporučeno.

Prirodoslovni muzej istražuje obrasce migracije dinosaura u Boliviji. Paleontolozi su otkrili otiske stopala dinosaura na N različitih nalazišta, numerirani od 0 do $N - 1$ prema opadajućem redoslijedu starosti: lokacija 0 sadrži najstarije otiske stopala, dok lokacija $N - 1$ sadrži najmlađe.

Dinosauri su migrirali na svako nalazište (osim nalazišta 0) s određenog, starijeg nalazišta. Za svako nalazište i takvo da je $1 \leq i \leq N - 1$, postoji točno jedno starije nalazište $P[i]$ (s $P[i] < i$) takvo da su neki dinosauri migrirali izravno s nalazišta $P[i]$ na nalazište i . Jedno starije nalazište moglo je biti izvor migracija na više mlađih nalazišta.

Paleontolozi modeliraju svaku migraciju kao **neusmjerenu vezu** između lokacija i i $P[i]$. Imajte na umu da je za bilo koje dvije različite lokacije x i y moguće putovati od x do y prolaskom kroz niz veza. **Udaljenost** između lokacija x i y definirana je kao minimalni broj veza potrebnih za putovanje od x do y .

Na primjer, sljedeća slika prikazuje slučaj s $N = 5$ lokacija, gdje $P[1] = 0$, $P[2] = 1$, $P[3] = 2$ i $P[4] = 2$. Od lokacije 3 do lokacije 4 može se putovati preko 2 presjedanja, tako da je udaljenost između njih 2.



Muzej ima za cilj identificirati par lokacija s najvećom mogućom udaljenošću.

Imajte na umu da par nije nužno jedinstven: na primjer, u gornjem primjeru, oba para $(0, 3)$ i $(0, 4)$ imaju udaljenost od 3, što je maksimalna udaljenost. U takvim slučajevima, bilo koji od parova koji postignu maksimalnu udaljenost smatra se valjanim.

U početku, vrijednosti $P[i]$ **nisu** poznate. Muzej šalje istraživački tim da redom posjeti lokalitete $1, 2, \dots, N - 1$. Nakon dolaska na svaku lokaciju i ($1 \leq i \leq N - 1$), tim izvršava obje sljedeće radnje:

- Određuje vrijednost $P[i]$, odnosno izvor migracije na lokaciju i .
- Odlučuje hoće li poslati **jednu** poruku Muzeju ili neće poslati nijednu poruku s ove stranice, na temelju prethodno prikupljenih informacija.

Poruke se prenose putem skupog satelitskog komunikacijskog sustava, tako da svaka poruka mora biti cijeli broj između 1 i 20 000, uključivo. Osim toga, timu je dopušteno poslati ukupno **najviše** 50

.

Vaš je zadatak provesti strategiju putem koje:

- Istraživački tim odabire stranice s kojih se šalju poruke, zajedno s vrijednošću svake poruke.
- Muzej može odrediti par lokacija s maksimalnom udaljenošću, isključivo na temelju poruka primljenih s svake lokacije.

Slanje velikih brojeva putem satelita je skupo. Vaš rezultat ovisit će i o najvećem poslanom cijelom broju i o ukupnom broju poslanih poruka.

Implementation Details

You need to implement two procedures; one for the research team, and another one for the Museum.

The procedure you should implement for the **research team** is:

```
int send_message(int N, int i, int Pi)
```

- N : the number of sites with footprints.
- i : the index of the site the team is currently visiting.
- Pi : the value of $P[i]$.
- This procedure is called $N - 1$ times for each test case, in the order of $i = 1, 2, \dots, N - 1$.

This procedure should return $S[i]$ specifying the action taken by the research team at site i :

- $S[i] = 0$: the research team decides not to send a message from site i .
- $1 \leq S[i] \leq 20\,000$: the research team sends the integer $S[i]$ as the message from site i .

The procedure you should implement for the **Museum** is:

```
std::pair<int,int> longest_path(std::vector<int> S)
```

- S : an array of length N such that:
 - $S[0] = 0$.
 - For each $1 \leq i \leq N - 1$, $S[i]$ is the integer returned by `send_message(N, i, Pi)`.
- This procedure is called exactly once for each test case.

This procedure should return a pair of sites (U, V) with the maximum distance.

In the actual grading, a program that calls the above procedures is run **exactly two times**.

- During the first run of the program:
 - `send_message` is called exactly $N - 1$ times.
 - **Your program can store and retain information across successive calls.**
 - The returned values (array S) are stored by the grading system.
 - In some cases the behavior of the grader is **adaptive**. This means that the value of $P[i]$ in a call to `send_message` may depend on the actions taken by the research team during the previous calls.
- During the second run of the program:
 - `longest_path` is called exactly once. The only information available to `longest_path` from the first run is the array S .

Constraints

- $N = 10\,000$
- $0 \leq P[i] < i$ for each i such that $1 \leq i \leq N - 1$.

Subtasks and Scoring

Subtask	Score	Additional Constraints
1	30	Site 0 and some other site have a maximum distance among all pairs of sites.
2	70	No additional constraints.

Let Z be the maximum integer appearing in the array S , and let M be the number of messages sent by the research team.

In any of the test cases, if at least one of the following conditions hold, then the score of your solution for that test case will be 0 (reported as `Output isn't correct` in CMS):

- At least one element in S is invalid.
- $Z > 20\,000$ or $M > 50$.
- The return value of the call to procedure `longest_path` is incorrect.

Otherwise, your score for subtask 1 is calculated as follows:

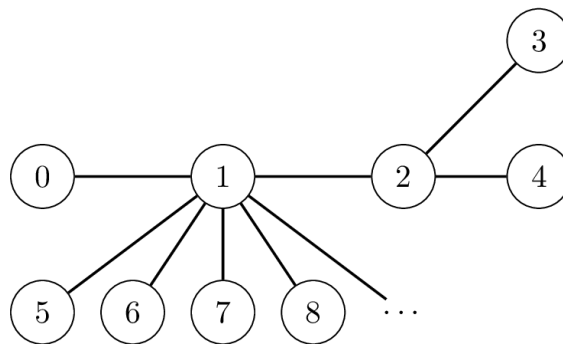
Condition	Score
$9\,998 \leq Z \leq 20\,000$	10
$102 \leq Z \leq 9997$	16
$5 \leq Z \leq 101$	23
$Z \leq 4$	30

Your score for subtask 2 is calculated as follows:

Condition	Score
$5 \leq Z \leq 20\,000$ and $M \leq 50$	$35 - 25 \log_{4000} \left(\frac{Z}{5} \right)$
$Z \leq 4$ and $32 \leq M \leq 50$	40
$Z \leq 4$ and $9 \leq M \leq 31$	$70 - 30 \log_4 \left(\frac{M}{8} \right)$
$Z \leq 4$ and $M \leq 8$	70

Example

Let $N = 10\,000$. Consider the situation in which $P[1] = 0$, $P[2] = 1$, $P[3] = 2$, $P[4] = 2$, and $P[i] = 1$ for $i > 4$.



Assume that the research team's strategy is to send the message $10 \cdot V + U$ whenever the pair of sites (U, V) with the maximum distance changes as a result of a `send_message` call.

Initially, the pair with the maximum distance is $(U, V) = (0, 0)$. Consider the following sequence of calls during the first run of the program:

Procedure call	(U, V)	Return value ($S[i]$)
<code>send_message(10000, 1, 0)</code>	$(0, 1)$	10
<code>send_message(10000, 2, 1)</code>	$(0, 2)$	20
<code>send_message(10000, 3, 2)</code>	$(0, 3)$	30
<code>send_message(10000, 4, 2)</code>	$(0, 3)$	0

Note that in all the remaining calls the value of $P[i]$ is 1. This means that the pair with the maximum distance does not change, and the team does not send any more messages.

Then, in the second run of the program, the following call is made:

```
longest_path([0, 10, 20, 30, 0, ...])
```

The Museum reads the last message sent by the research team, which is $S[3] = 30$, and deduces that $(0, 3)$ is the pair of sites with the maximum distance. Therefore, the call returns $(0, 3)$.

Note that this approach does not always make it possible for the Museum to determine the pair with the maximum distance correctly.

Sample Grader

The sample grader calls both `send_message` and `longest_path` in the same run, which is different from the actual grader.

Input format:

```
N
P[1] P[2] ... P[N-1]
```

Output format:

```
S[1] S[2] ... S[N-1]
U V
```

Note that you can use the sample grader with any value of N .